

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Thursday 12 March 2026

1.1 Daily Limit

Exponential Limit with Trigonometric Difference

Evaluate the limit:

$$\lim_{x \rightarrow 0} e^{\frac{\tan(x) - \sin(x)}{x^3}}$$

Solution: Let us first evaluate the limit of the exponent. We can use the Maclaurin series expansions for the tangent and sine functions near $x = 0$:

$$\tan(x) = x + \frac{x^3}{3} + O(x^5)$$

$$\sin(x) = x - \frac{x^3}{3!} + O(x^5) = x - \frac{x^3}{6} + O(x^5)$$

Subtracting the sine series from the tangent series gives:

$$\tan(x) - \sin(x) = \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) + O(x^5) = \frac{x^3}{3} + \frac{x^3}{6} + O(x^5) = \frac{x^3}{2} + O(x^5)$$

Now, substituting this back into the exponent's expression:

$$\lim_{x \rightarrow 0} \frac{\tan(x) - \sin(x)}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{x^3}{2} + O(x^5)}{x^3} = \frac{1}{2}$$

Since the exponential function $f(u) = e^u$ is continuous, we can pass the limit inside:

$$\lim_{x \rightarrow 0} e^{\frac{\tan(x) - \sin(x)}{x^3}} = e^{1/2} = \sqrt{e}$$

Friday 13 March 2026

2.1 Daily Limit

Limit of a Sum Involving a Derivative Definition

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \sum_{n=1}^x \lim_{k \rightarrow 0} \frac{(\sin(n/x) \cos(k) + \sin(k) \cos(n/x))^{\sin(n/x) \cos(k) + \sin(k) \cos(n/x)} - (\sin(n/x))^{\sin(n/x)}}{kx}$$

Solution: First, recognize the angle addition formula in the numerator: $\sin(A + B) = \sin(A) \cos(B) + \sin(B) \cos(A)$. Setting $A = n/x$ and $B = k$, the base and exponent simplify to $\sin(n/x + k)$. Let $u = n/x$. The inner limit with respect to k becomes the definition of the derivative evaluated at $k = 0$. We can rewrite the expression as:

$$\frac{1}{x} \lim_{k \rightarrow 0} \frac{(\sin(u + k))^{\sin(u + k)} - (\sin(u))^{\sin(u)}}{k}$$

This limit is exactly the derivative of $f(y) = y^y$ with respect to k , where $y = \sin(u + k)$, evaluated at $k = 0$. Using the chain rule, $\frac{d}{dk}[(\sin(u + k))^{\sin(u + k)}] = \frac{d}{dy}(y^y) \cdot \frac{dy}{dk}$. The derivative of y^y is $y^y(1 + \ln y)$. Thus, at $k = 0$, the derivative is:

$$(\sin(u))^{\sin(u)}(1 + \ln(\sin u)) \cos(u)$$

Substituting this back into our original limit yields a Riemann sum:

$$L = \lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n=1}^x (\sin(n/x))^{\sin(n/x)} (1 + \ln(\sin(n/x))) \cos(n/x)$$

As $x \rightarrow \infty$, this sum converges to the definite integral over $[0, 1]$:

$$L = \int_0^1 (\sin u)^{\sin u} (1 + \ln(\sin u)) \cos(u) du$$

Let $w = \sin u$, which gives $dw = \cos(u) du$. The bounds change from $[0, 1]$ to $[0, \sin(1)]$:

$$L = \int_0^{\sin(1)} w^w (1 + \ln w) dw$$

Recognizing that $w^w(1 + \ln w)$ is the exact derivative of w^w , we evaluate the anti-derivative:

$$L = \left[w^w \right]_0^{\sin(1)}$$

Taking the limit as $w \rightarrow 0^+$, $w^w \rightarrow 1$. Therefore:

$$L = (\sin 1)^{\sin 1} - 1$$

Saturday 14 March 2026

3.1 Daily Limit

Gamma Function and Exponential Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0^+} \left(\frac{\Gamma\left(\frac{1}{x} + 1 + x\right)}{\Gamma\left(\frac{1}{x} + 1\right)} x^x \right)^{\frac{1}{x^2}}$$

Solution: Let $z = 1/x$. As $x \rightarrow 0^+$, we have $z \rightarrow \infty$. Rewriting the limit in terms of z :

$$L = \lim_{z \rightarrow \infty} \left(\frac{\Gamma(z + 1 + 1/z)}{\Gamma(z + 1)} z^{-1/z} \right)^{z^2}$$

We take the natural logarithm of the expression, denoted as $\ln L$, which transforms the exponent into a multiplier:

$$\ln L = \lim_{z \rightarrow \infty} z^2 \left(\ln \Gamma\left(z + 1 + \frac{1}{z}\right) - \ln \Gamma(z + 1) - \frac{\ln z}{z} \right)$$

We use the Taylor expansion of the log-gamma function around $Z = z + 1$ with a small perturbation $a = 1/z$:

$$\ln \Gamma(Z + a) - \ln \Gamma(Z) = a\psi(Z) + \frac{a^2}{2}\psi'(Z) + \frac{a^3}{6}\psi''(Z) + O(a^4)$$

Using the asymptotic expansions of the polygamma functions as $z \rightarrow \infty$:

$$\psi(z + 1) = \ln z + \frac{1}{2z} - \frac{1}{12z^2} + O\left(\frac{1}{z^4}\right)$$

$$\psi'(z + 1) = \frac{1}{z} - \frac{1}{2z^2} + O\left(\frac{1}{z^3}\right)$$

$$\psi''(z + 1) = -\frac{1}{z^2} + O\left(\frac{1}{z^3}\right)$$

Substituting $a = 1/z$ and these expansions back into the Taylor series:

$$\ln \Gamma\left(z + 1 + \frac{1}{z}\right) - \ln \Gamma(z + 1) = \frac{1}{z} \left(\ln z + \frac{1}{2z} - \frac{1}{12z^2} \right) + \frac{1}{2z^2} \left(\frac{1}{z} \right) + \frac{1}{6z^3} \left(-\frac{1}{z^2} \right) + O\left(\frac{1}{z^4}\right)$$

Simplifying this expression:

$$\frac{\ln z}{z} + \frac{1}{2z^2} - \frac{1}{12z^3} + \frac{1}{2z^3} + O\left(\frac{1}{z^4}\right) = \frac{\ln z}{z} + \frac{1}{2z^2} + \frac{5}{12z^3} + O\left(\frac{1}{z^4}\right)$$

Substitute this back into the log-limit equation:

$$\ln L = \lim_{z \rightarrow \infty} z^2 \left(\frac{\ln z}{z} + \frac{1}{2z^2} + \frac{5}{12z^3} - \frac{\ln z}{z} + O\left(\frac{1}{z^4}\right) \right)$$

The $\frac{\ln z}{z}$ terms cancel out, leaving:

$$\ln L = \lim_{z \rightarrow \infty} z^2 \left(\frac{1}{2z^2} + \frac{5}{12z^3} \right) = \lim_{z \rightarrow \infty} \left(\frac{1}{2} + \frac{5}{12z} \right) = \frac{1}{2}$$

Exponentiating both sides yields the final result:

$$L = e^{1/2} = \sqrt{e}$$

Sunday 15 March 2026

4.1 Daily Limit

Limit of a Ratio of Sums

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{\frac{n}{1} + \frac{n}{\sqrt{2}} + \frac{n}{\sqrt{3}} + \cdots + \frac{n}{\sqrt{n}}}{1 + \sqrt{2} + \sqrt{3} + \cdots + \sqrt{n}}$$

Solution: We can rewrite the limit using summation notation:

$$L = \lim_{n \rightarrow \infty} \frac{n \sum_{k=1}^n \frac{1}{\sqrt{k}}}{\sum_{k=1}^n \sqrt{k}}$$

To evaluate the asymptotic behavior of the numerator and the denominator, we manipulate both sums into Riemann sums by factoring out appropriate powers of n . For the numerator:

$$N_n = n \sum_{k=1}^n \frac{1}{\sqrt{k}} = n \cdot \sqrt{n} \sum_{k=1}^n \frac{1}{n} \frac{1}{\sqrt{k/n}} = n^{3/2} \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n} \right)^{-1/2}$$

For the denominator:

$$D_n = \sum_{k=1}^n \sqrt{k} = \sqrt{n} \cdot n \sum_{k=1}^n \frac{1}{n} \sqrt{k/n} = n^{3/2} \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n} \right)^{1/2}$$

Substituting these back into the limit:

$$L = \lim_{n \rightarrow \infty} \frac{n^{3/2} \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n} \right)^{-1/2}}{n^{3/2} \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n} \right)^{1/2}}$$

The $n^{3/2}$ terms cancel out. As $n \rightarrow \infty$, the summations converge to definite integrals over the interval $[0, 1]$:

$$L = \frac{\int_0^1 x^{-1/2} dx}{\int_0^1 x^{1/2} dx}$$

We evaluate the integrals individually:

$$\int_0^1 x^{-1/2} dx = \left[2x^{1/2} \right]_0^1 = 2$$

$$\int_0^1 x^{1/2} dx = \left[\frac{2}{3} x^{3/2} \right]_0^1 = \frac{2}{3}$$

Dividing the results gives the final value:

$$L = \frac{2}{2/3} = 3$$

Monday 16 March 2026

5.1 Daily Limit

Exponential Sum Limit via Integration

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{ne^{-\ln^2(\frac{k}{n-k})}}{k(n-k)}$$

Solution: We begin by expressing the summand in a form suitable for a Riemann sum. Factoring out n^2 from the denominator:

$$k(n-k) = n^2 \left(\frac{k}{n}\right) \left(1 - \frac{k}{n}\right)$$

Substitute this back into the limit:

$$L = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{ne^{-\ln^2(\frac{k/n}{1-k/n})}}{n^2 \left(\frac{k}{n}\right) \left(1 - \frac{k}{n}\right)} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{e^{-\ln^2(\frac{k/n}{1-k/n})}}{\left(\frac{k}{n}\right) \left(1 - \frac{k}{n}\right)}$$

Let $x = k/n$ with $\Delta x = 1/n$. As $n \rightarrow \infty$, this sum converges to the definite integral over $[0, 1]$:

$$L = \int_0^1 \frac{1}{x(1-x)} e^{-\ln^2(\frac{x}{1-x})} dx$$

To evaluate this integral, we use the substitution $u = \ln\left(\frac{x}{1-x}\right)$. Differentiating u with respect to x gives:

$$du = \left(\frac{1}{x} + \frac{1}{1-x}\right) dx = \frac{1}{x(1-x)} dx$$

This perfectly matches the algebraic fraction in our integrand. Next, we determine the new limits of integration. As $x \rightarrow 0^+$, the term $\frac{x}{1-x} \rightarrow 0^+$, so $u \rightarrow -\infty$. As $x \rightarrow 1^-$, the term $\frac{x}{1-x} \rightarrow \infty$, so $u \rightarrow \infty$. The integral simplifies elegantly to the standard Gaussian integral:

$$L = \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$$

Tuesday 17 March 2026

6.1 Daily Limit

Asymptotic Expansion of the Central Binomial Coefficient

Evaluate the limit:

$$\lim_{x \rightarrow \infty} x \left(\frac{\sqrt{\pi x} \binom{2x}{x}}{4^x} - 1 \right)$$

Solution: To resolve this limit, we need the precise asymptotic expansion of the central binomial coefficient $\binom{2x}{x}$ for large x . We utilize Stirling's approximation for factorials, which states:

$$N! \sim \sqrt{2\pi N} \left(\frac{N}{e} \right)^N \left(1 + \frac{1}{12N} + \frac{1}{288N^2} - \dots \right)$$

Applying this to the components of the central binomial coefficient $\binom{2x}{x} = \frac{(2x)!}{(x!)^2}$: For the numerator:

$$(2x)! \sim \sqrt{4\pi x} \left(\frac{2x}{e} \right)^{2x} \left(1 + \frac{1}{24x} + \dots \right)$$

For the denominator squared:

$$(x!)^2 \sim \left(\sqrt{2\pi x} \left(\frac{x}{e} \right)^x \left(1 + \frac{1}{12x} + \dots \right) \right)^2 \sim 2\pi x \left(\frac{x}{e} \right)^{2x} \left(1 + \frac{1}{6x} + \dots \right)$$

Dividing the two yields the asymptotic expansion of $\binom{2x}{x}$:

$$\binom{2x}{x} \sim \frac{\sqrt{4\pi x} 2^{2x} x^{2x} / e^{2x}}{2\pi x \frac{x^{2x}}{e^{2x}} \left(1 + \frac{1}{6x} \right)} \sim \frac{4^x}{\sqrt{\pi x}} \left(1 + \frac{1}{24x} \right) \left(1 - \frac{1}{6x} \right)$$

Multiplying the Taylor polynomials up to the $1/x$ term gives:

$$\binom{2x}{x} \sim \frac{4^x}{\sqrt{\pi x}} \left(1 + \frac{1}{24x} - \frac{1}{6x} + O\left(\frac{1}{x^2}\right) \right) = \frac{4^x}{\sqrt{\pi x}} \left(1 - \frac{3}{24x} + O\left(\frac{1}{x^2}\right) \right) = \frac{4^x}{\sqrt{\pi x}} \left(1 - \frac{1}{8x} + O\left(\frac{1}{x^2}\right) \right)$$

Substitute this expansion back into the original limit expression:

$$L = \lim_{x \rightarrow \infty} x \left(\frac{\sqrt{\pi x}}{4^x} \left[\frac{4^x}{\sqrt{\pi x}} \left(1 - \frac{1}{8x} + O\left(\frac{1}{x^2}\right) \right) \right] - 1 \right)$$

$$L = \lim_{x \rightarrow \infty} x \left(1 - \frac{1}{8x} + O\left(\frac{1}{x^2}\right) - 1 \right) = \lim_{x \rightarrow \infty} x \left(-\frac{1}{8x} + O\left(\frac{1}{x^2}\right) \right)$$

Taking the limit as $x \rightarrow \infty$:

$$L = -\frac{1}{8}$$

Wednesday 18 March 2026

7.1 Daily Limit

Limit Involving Error Function and Sine Integral

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{\operatorname{erf}(1/n)}{\operatorname{Si}(1/n)}$$

Solution: Let us perform a change of variables by setting $x = 1/n$. As $n \rightarrow \infty$, $x \rightarrow 0$. The limit can be rewritten as:

$$L = \lim_{x \rightarrow 0} \frac{\operatorname{erf}(x)}{\operatorname{Si}(x)}$$

Both the error function $\operatorname{erf}(x)$ and the Sine integral $\operatorname{Si}(x)$ evaluate to 0 at $x = 0$, producing an indeterminate form of type $\frac{0}{0}$. We apply L'Hôpital's Rule:

$$L = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \operatorname{erf}(x)}{\frac{d}{dx} \operatorname{Si}(x)}$$

The derivative of the error function is derived from its integral definition $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$:

$$\frac{d}{dx} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$$

Similarly, the derivative of the Sine integral $\operatorname{Si}(x) = \int_0^x \frac{\sin t}{t} dt$ is:

$$\frac{d}{dx} \operatorname{Si}(x) = \frac{\sin x}{x}$$

Substitute the derivatives back into the limit:

$$L = \lim_{x \rightarrow 0} \frac{\frac{2}{\sqrt{\pi}} e^{-x^2}}{\frac{\sin x}{x}}$$

We can evaluate this limit directly using standard limits. As $x \rightarrow 0$, $e^{-x^2} \rightarrow e^0 = 1$ and $\frac{\sin x}{x} \rightarrow 1$:

$$L = \frac{\frac{2}{\sqrt{\pi}}(1)}{1} = \frac{2}{\sqrt{\pi}}$$

Thursday 19 March 2026

8.1 Daily Limit

Limit of Subfactorials and Factorials

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{!(n^2)}{(n^2)! + (n+1)!}$$

Solution: The notation $!N$ denotes the subfactorial (or derangement number) of N . The subfactorial can be expressed in terms of the factorial using the well-known formula:

$$!N = N! \sum_{k=0}^N \frac{(-1)^k}{k!}$$

As $N \rightarrow \infty$, the summation represents the Taylor series expansion of e^x evaluated at $x = -1$. Hence, the asymptotic equivalence for the subfactorial is:

$$!N \sim \frac{N!}{e} \quad \text{as } N \rightarrow \infty$$

In our problem, $N = n^2$. Thus, the numerator exhibits the asymptotic behavior:

$$!(n^2) \sim \frac{(n^2)!}{e}$$

Now let us analyze the denominator: $(n^2)! + (n+1)!$. We factor out the dominant term $(n^2)!$:

$$(n^2)! + (n+1)! = (n^2)! \left(1 + \frac{(n+1)!}{(n^2)!} \right)$$

Since n^2 grows much faster than $n+1$, for $n \geq 2$, $n^2 > n+1$. Thus, as $n \rightarrow \infty$, the ratio $\frac{(n+1)!}{(n^2)!} \rightarrow 0$. The denominator is asymptotically equivalent to $(n^2)!$. Substituting these asymptotic behaviors back into the original limit:

$$L = \lim_{n \rightarrow \infty} \frac{\frac{(n^2)!}{e}}{(n^2)! \left(1 + \frac{(n+1)!}{(n^2)!} \right)}$$

Canceling $(n^2)!$ from the numerator and the denominator:

$$L = \lim_{n \rightarrow \infty} \frac{1/e}{1+0} = \frac{1}{e}$$

Friday 20 March 2026

9.1 Daily Limit

Infinite Series Limit with Quadratic Denominator

Evaluate the limit:

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{100n^2 - 100n + 9}$$

Solution: First, we factor the quadratic expression in the denominator:

$$100n^2 - 100n + 9 = (10n - 9)(10n - 1)$$

The sum becomes a rational series that we can decompose using partial fractions:

$$\frac{1}{(10n - 9)(10n - 1)} = \frac{A}{10n - 9} + \frac{B}{10n - 1}$$

Solving for the constants A and B , we find $A = 1/8$ and $B = -1/8$. Thus:

$$L = \frac{1}{8} \sum_{n=1}^{\infty} \left(\frac{1}{10n - 9} - \frac{1}{10n - 1} \right)$$

To evaluate this infinite sum, we can rewrite it using the Digamma function $\psi(z)$, which is related to infinite series by the identity $\sum_{n=0}^{\infty} \left(\frac{1}{n+a} - \frac{1}{n+b} \right) = \psi(b) - \psi(a)$. We factor out 10 from the denominators to match the standard form:

$$L = \frac{1}{8} \sum_{n=1}^{\infty} \left(\frac{1}{10(n - 9/10)} - \frac{1}{10(n - 1/10)} \right) = \frac{1}{80} \sum_{n=0}^{\infty} \left(\frac{1}{n + 1/10} - \frac{1}{n + 9/10} \right)$$

Here, $a = 1/10$ and $b = 9/10$. Applying the Digamma identity:

$$L = \frac{1}{80} \left(\psi(9/10) - \psi(1/10) \right)$$

We use the Euler reflection formula for the Digamma function, $\psi(1 - x) - \psi(x) = \pi \cot(\pi x)$, with $x = 1/10$:

$$\psi(9/10) - \psi(1/10) = \pi \cot\left(\frac{\pi}{10}\right)$$

Substituting this back into the limit expression gives the final exact form:

$$L = \frac{\pi}{80} \cot\left(\frac{\pi}{10}\right) = \frac{\pi \sqrt{5 + 2\sqrt{5}}}{80}$$

Saturday 21 March 2026

10.1 Daily Limit

Nested Radical and Trigonometric Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\left(\sqrt{1 - 4x \sqrt{1 - 4x \sqrt{\dots}}} \right) \left(\tan \left(\frac{\pi}{2} - x \right) \right)}{\Gamma(x)}$$

Solution: Let $y(x)$ denote the nested radical in the numerator:

$$y(x) = \sqrt{1 - 4x \sqrt{1 - 4x \sqrt{\dots}}}$$

By the self-similar nature of the infinite nested radical, we can write:

$$y(x) = \sqrt{1 - 4xy(x)}$$

Squaring both sides and rearranging yields a quadratic equation in terms of $y(x)$:

$$y(x)^2 + 4xy(x) - 1 = 0$$

Applying the quadratic formula to solve for $y(x)$:

$$y(x) = \frac{-4x \pm \sqrt{16x^2 - 4(1)(-1)}}{2} = -2x \pm \sqrt{4x^2 + 1}$$

Since the nested radical implies a positive square root and must equal 1 as $x \rightarrow 0$, we select the positive branch:

$$y(x) = \sqrt{4x^2 + 1} - 2x$$

As $x \rightarrow 0$, $y(x) \approx 1 - 2x + \dots \rightarrow 1$. The second term in the numerator simplifies via complementary angles:

$$\tan \left(\frac{\pi}{2} - x \right) = \cot(x)$$

As $x \rightarrow 0$, the cotangent function behaves asymptotically as $\cot(x) \sim \frac{1}{x}$. For the denominator, the Gamma function has a well-known simple pole at $x = 0$, with the asymptotic behavior:

$$\Gamma(x) \sim \frac{1}{x} \quad \text{as } x \rightarrow 0$$

Now, substitute these leading-order asymptotic behaviors back into the original limit quotient:

$$L = \lim_{x \rightarrow 0} \frac{y(x) \cot(x)}{\Gamma(x)} = \lim_{x \rightarrow 0} \frac{(1) \left(\frac{1}{x} \right)}{\left(\frac{1}{x} \right)}$$

The singular $1/x$ terms cancel out precisely, yielding:

$$L = 1$$